



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO KAMALNATH FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

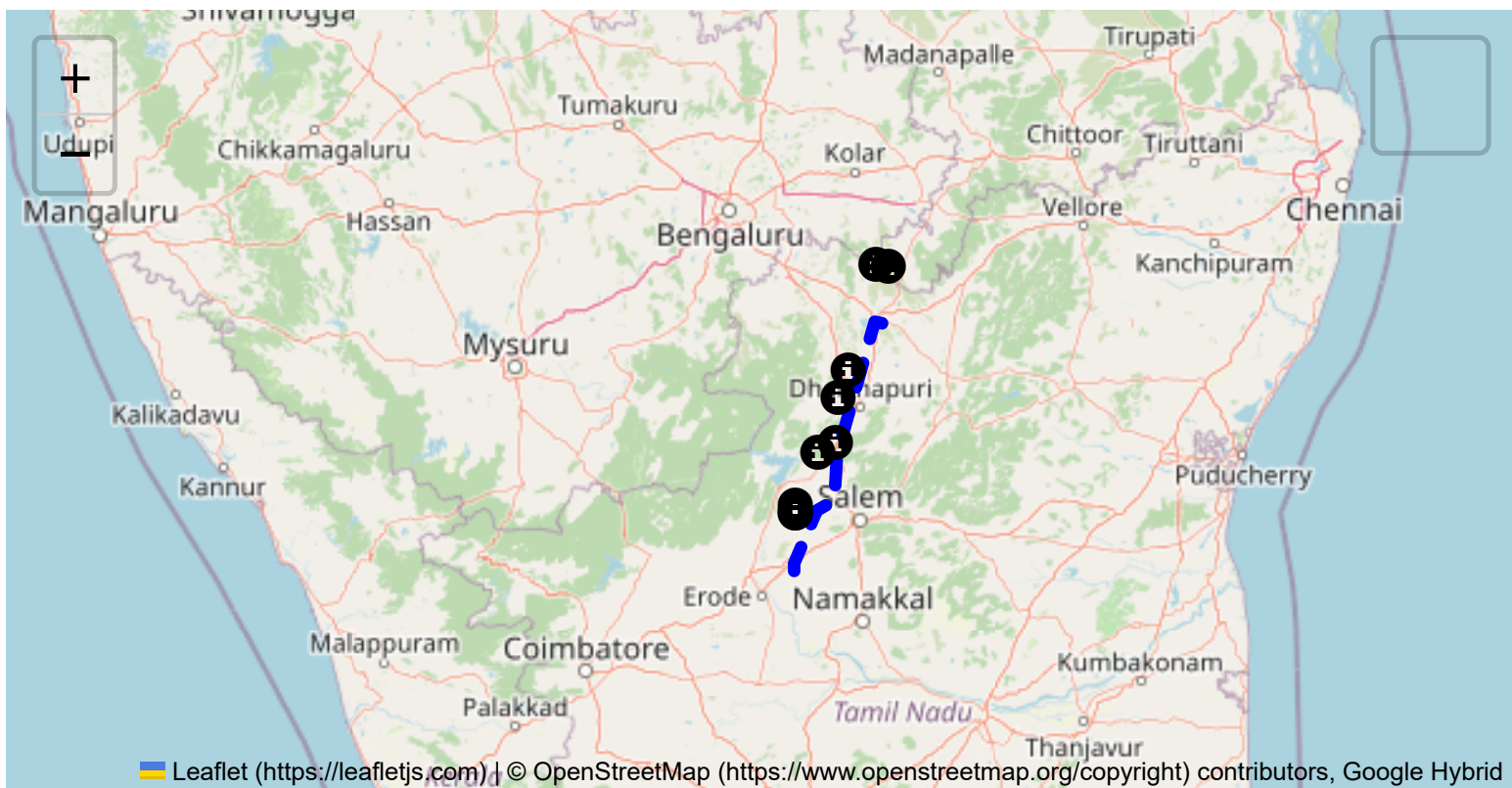
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

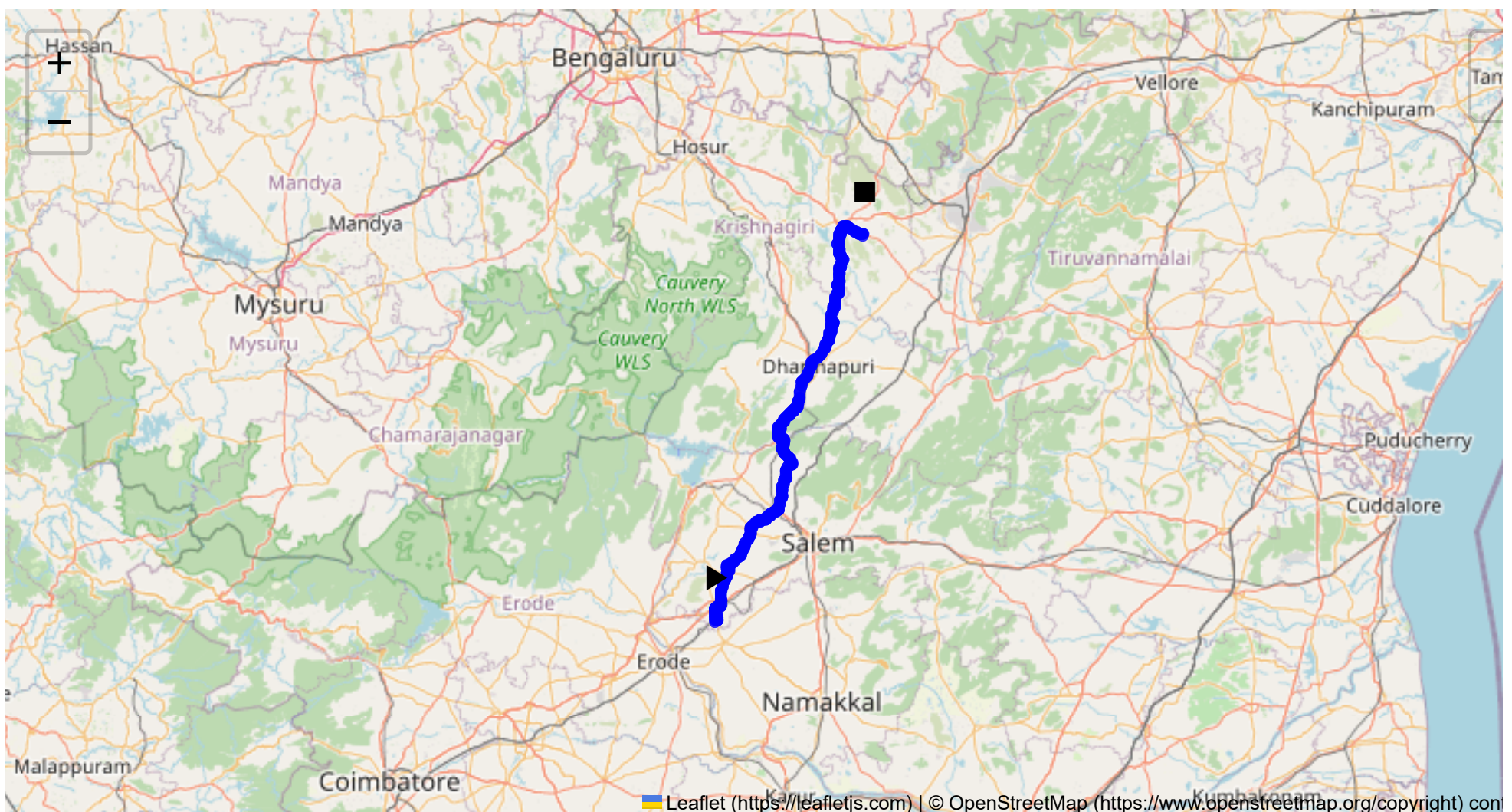
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 147.96 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.492564, 78.282379)



Welcome to the Journey Risk Management Study

Analyzing the proposed route from CVQF+23W, Sangagiri, Tamil Nadu to F7VJ+5V, Chinnapanamudlu, Tamil Nadu, here's a detailed safety assessment:

1. Overview of the Route Map

- The route spans roughly 148 kilometers through key towns and areas in Tamil Nadu, including Sangagiri, Kangeyampalayam, Salem, Tharamangalam, Omalur, Thoppur, Seshampatti, and

Krishnagiri. It involves traveling primarily via state highways, which are crucial for regional connectivity.

2. Typical Weather Conditions and Hazards

- The region experiences tropical weather with distinct dry and wet seasons. Monsoons (June to September) can lead to heavy rainfall, causing waterlogging and reduced visibility, which are key concerns for heavy vehicle navigation.
- During summer months, high temperatures may lead to road surface damage.

3. Traffic Patterns & Congestion

- Peak traffic hours generally occur between 8:00-10:00 AM and 5:00-7:00 PM. Salem and Krishnagiri, being major junctions, tend to be congested during these times.
- Congestion-prone areas include highway intersections, toll plazas, and city centers.

4. Road Quality and Infrastructure

- Overall, state highways are well-maintained, but some segments may have potholes or uneven surfaces, particularly post-monsoon.
- Watch for roadworks or extensions near rapidly developing areas like Salem.

5. Alternative Routes for Emergencies

- In case of severe congestion or incidents, the NH 44 via Salem can serve as a broader highway route, although it may be longer in terms of distance.
- Local bypasses around urban centers can provide temporary relief from congestion.

6. Local Regulations for Hazardous Material Transport

- Transporting hazardous materials requires compliance with specific state and national regulations, including display of hazard placards and adherence to designated routes where heavy vehicle traffic is permitted.
- Night travel may be restricted in certain areas, so check local ordinances.

7. Historical Incidents Involving Heavy Vehicles

- There have been reports of road accidents involving heavy vehicles in this region, typically due to driver fatigue, brake failures on slopes, or misjudgment in busy areas.
- The Salem to Dharmapuri stretch has witnessed incidents related to overspeeding.

8. Environmental Considerations and Sensitive Areas

- Certain sections near Thoppur are ecologically sensitive and part of forest ranges. Be alert to wildlife crossings and ensure that hazardous material transport follows governmental guidelines.
- Avoid disposing of any waste materials along the route, particularly near water bodies.

9. Communication Coverage and Potential Dead Zones

- The route generally has good communication coverage, but certain hilly areas near Dharmapuri might experience weak signals.
- Familiarize drivers with alternative communication methods like radio or satellite phones in these areas.

10. Estimated Emergency Response Times

- Urban centers such as Salem and Krishnagiri have quicker response times (approximately 15-30 minutes), while more remote sections of the route could experience delays of up to an hour.
- Emergency services are generally well-organized in towns, with hospitals and firefighting units accessible within major towns.

11. Overall Summary of Risk Assessment

- This route is generally safe for transport of hazardous materials, assuming adherence to regulations and cautious driving, especially during adverse weather or peak traffic.
- Attention to potential roadworks, weather conditions, and congestion hotspots is crucial.
- Proactive route planning and regular communication can mitigate most risks, ensuring a smooth journey for truck drivers.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.4749755, 77.869149	10 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44028, 77.87541	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	Medium	11.46392, 77.87328	30 KM/Hr
6	Turn	High	11.47391, 77.86844	15 KM/Hr
7	Blind Spot	Blind Spot	11.47481, 77.86920	10 KM/Hr
8	Turn	High	11.47464, 77.86938	15 KM/Hr
9	Turn	Medium	11.48005, 77.88450	30 KM/Hr
10	Turn	Medium	11.48385, 77.88782	30 KM/Hr
11	Turn	Medium	11.49155, 77.88776	30 KM/Hr
12	Turn	High	11.49469, 77.88488	15 KM/Hr
13	Turn	High	11.55614, 77.89764	15 KM/Hr
14	Turn	Medium	11.55680, 77.89969	30 KM/Hr
15	Turn	High	11.58218, 77.90845	15 KM/Hr
16	Turn	High	11.58330, 77.90642	15 KM/Hr
17	Turn	Medium	11.59189, 77.90645	30 KM/Hr
18	Turn	Medium	11.59328, 77.90809	30 KM/Hr
19	Turn	Medium	11.63867, 77.94841	30 KM/Hr
20	Turn	High	11.64217, 77.95200	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
21	Turn	Medium	11.64712, 77.95118	30 KM/Hr
22	Turn	High	11.69597, 77.97126	15 KM/Hr
23	Turn	Medium	11.69794, 77.97520	30 KM/Hr
24	Turn	Medium	11.71380, 77.99532	30 KM/Hr
25	Turn	Medium	11.71731, 78.00006	30 KM/Hr
26	Turn	Medium	11.71947, 78.01172	30 KM/Hr
27	Turn	Medium	11.72472, 78.01630	30 KM/Hr
28	Turn	Medium	11.72760, 78.02285	30 KM/Hr
29	Turn	Medium	11.73761, 78.03437	30 KM/Hr
30	Turn	Medium	11.73759, 78.03658	30 KM/Hr
31	Turn	High	11.73652, 78.03863	15 KM/Hr
32	Turn	Medium	11.74252, 78.04705	30 KM/Hr
33	Turn	Medium	11.77968, 78.05777	30 KM/Hr
34	Turn	Medium	11.86887, 78.08377	30 KM/Hr
35	Turn	Medium	11.87976, 78.08182	30 KM/Hr
36	Turn	Medium	11.92621, 78.06514	30 KM/Hr
37	Turn	Medium	11.93719, 78.05372	30 KM/Hr
38	Turn	High	11.95828, 78.04983	15 KM/Hr
39	Turn	Medium	11.95997, 78.04946	30 KM/Hr
40	Turn	Medium	11.96335, 78.05141	30 KM/Hr
41	Turn	High	11.96408, 78.05131	15 KM/Hr
42	Turn	Medium	11.96511, 78.04966	30 KM/Hr
43	Turn	High	11.96608, 78.04919	15 KM/Hr
44	Turn	Medium	11.96819, 78.05088	30 KM/Hr
45	Turn	Medium	12.16993, 78.16780	30 KM/Hr
46	Turn	Medium	12.19885, 78.18158	30 KM/Hr
47	Turn	Medium	12.22467, 78.19138	30 KM/Hr
48	Turn	Medium	12.24611, 78.19519	30 KM/Hr
49	Turn	Medium	12.25061, 78.19978	30 KM/Hr
50	Turn	Medium	12.25349, 78.20037	30 KM/Hr
51	Turn	Medium	12.29134, 78.19920	30 KM/Hr
52	Turn	Medium	12.40597, 78.22280	30 KM/Hr
53	Turn	Medium	12.45026, 78.21827	30 KM/Hr
54	Turn	High	12.50709, 78.22230	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
55	Turn	High	12.51556, 78.23049	15 KM/Hr
56	Turn	Medium	12.51540, 78.24044	30 KM/Hr
57	Turn	Medium	12.49340, 78.28179	30 KM/Hr
58	Turn	Medium	12.49293, 78.28194	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
10	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
11	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
12	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
13	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
14	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
15	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
16	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
17	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
18	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
19	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
21	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
22	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
23	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
24	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
25	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
27	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
28	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
29	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
30	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
31	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
32	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
33	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
35	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
37	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
41	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
46	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
20	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
34	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
36	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
40	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
42	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
43	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
44	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
45	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.4749755, 77.869149



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46392, 77.87328



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47391, 77.86844

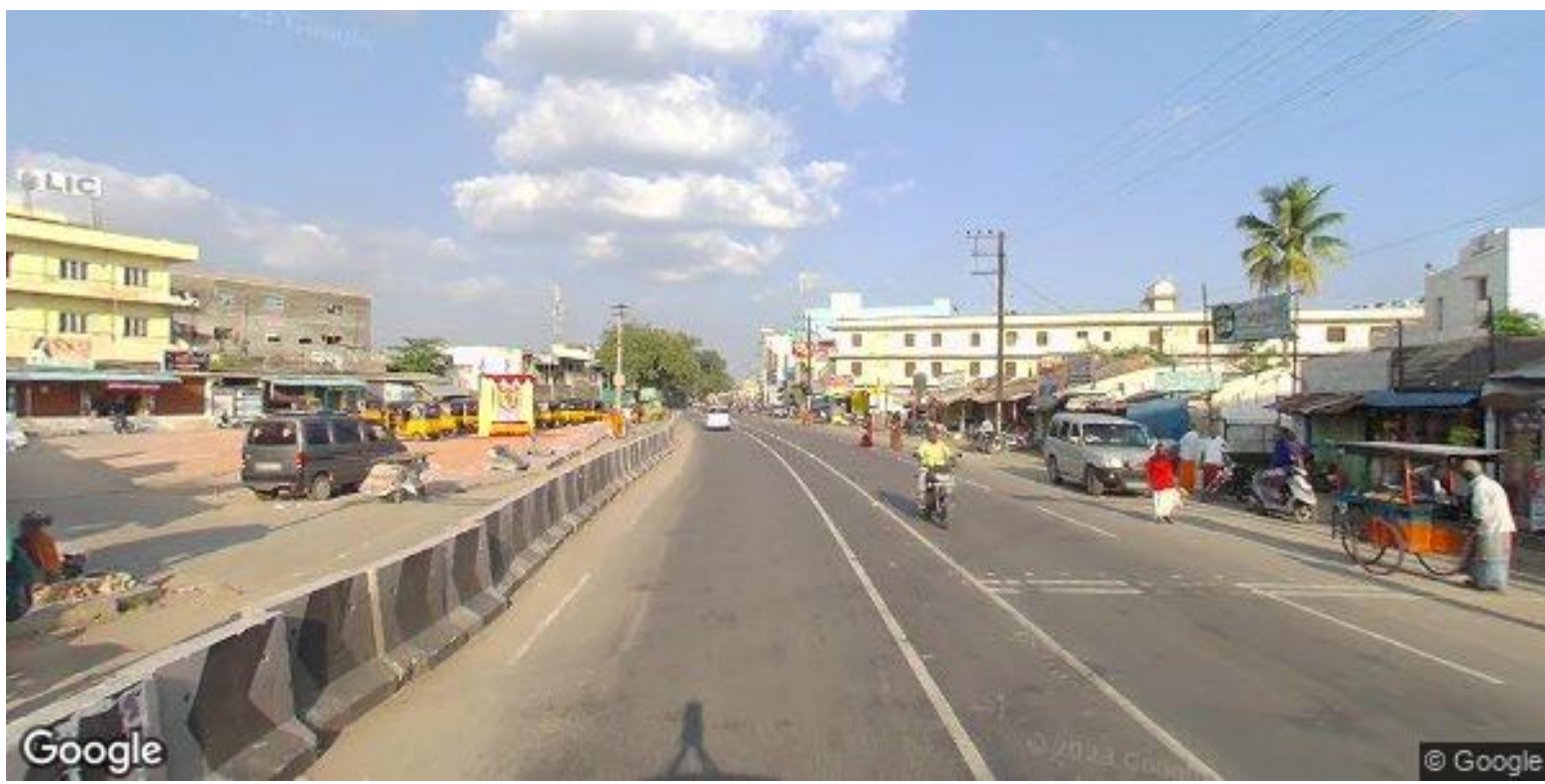


Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.47481, 77.86920



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47464, 77.86938



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48005, 77.88450



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48385, 77.88782



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49155, 77.88776



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49469, 77.88488

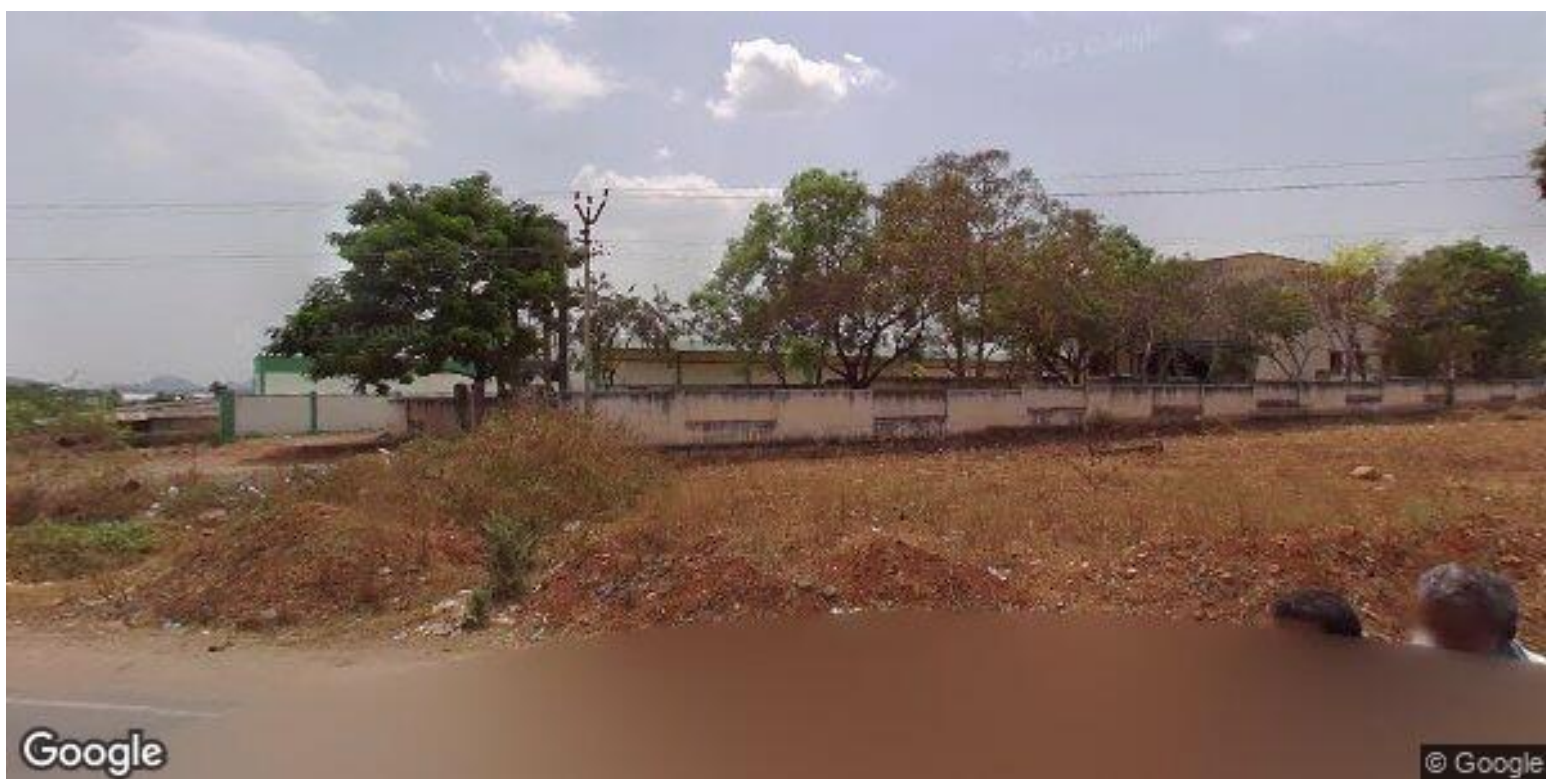


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55614, 77.89764



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58330, 77.90642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69597, 77.97126



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69794, 77.97520



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71380, 77.99532



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71731, 78.00006



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71947, 78.01172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72472, 78.01630



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72760, 78.02285



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73761, 78.03437



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73759, 78.03658



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03863



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74252, 78.04705



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86887, 78.08377



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87976, 78.08182



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.92621, 78.06514

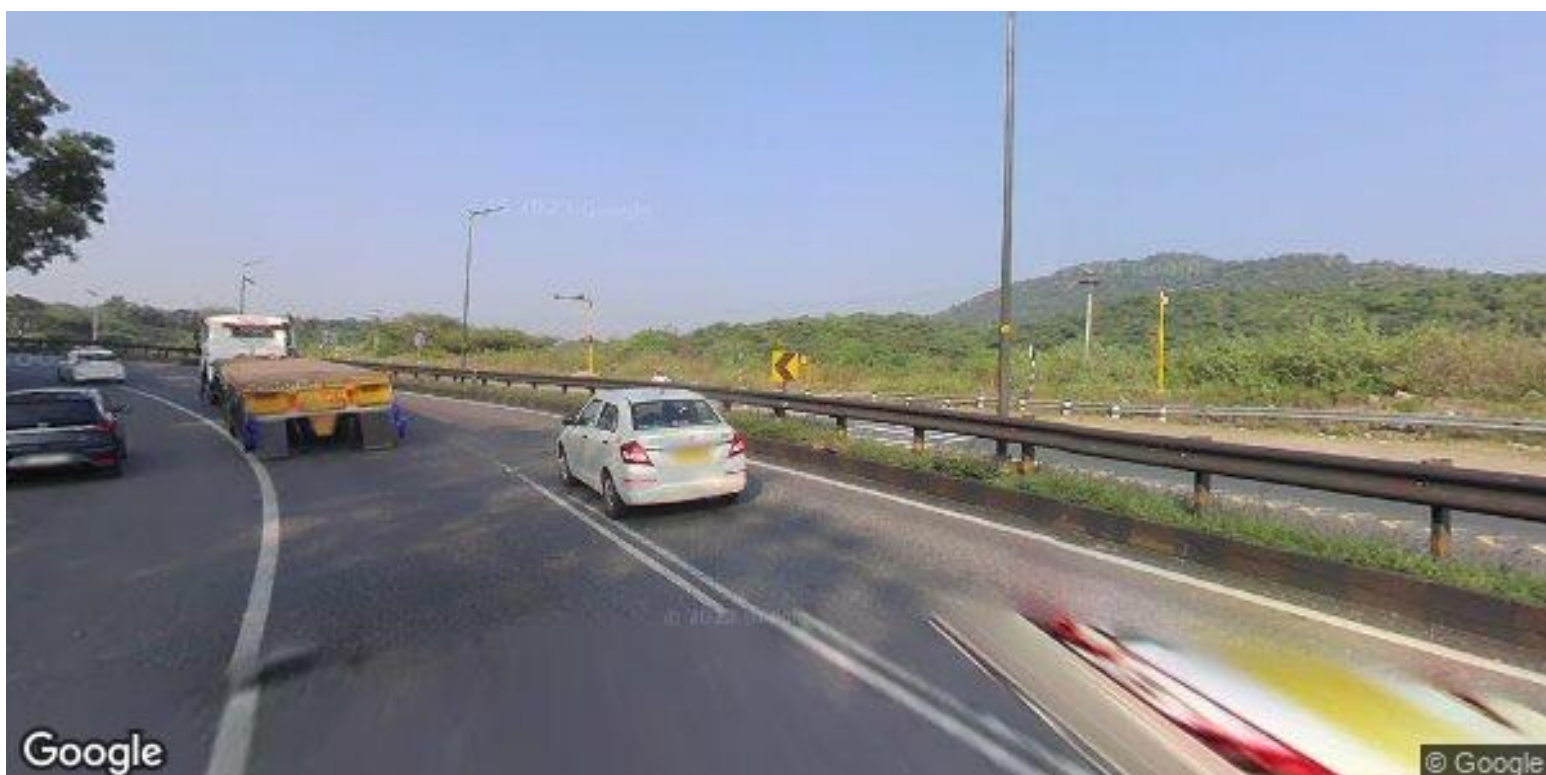


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95997, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96335, 78.05141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96408, 78.05131



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16993, 78.16780



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.19885, 78.18158



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22467, 78.19138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24611, 78.19519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25061, 78.19978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25349, 78.20037



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29134, 78.19920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40597, 78.22280



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.45026, 78.21827



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.50709, 78.22230



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51540, 78.24044



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.49340, 78.28179



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.49293, 78.28194
